Available online at www.sciencedirect.com
SCIENCE DIRECT®
ELSEVIER
Proceedings of the Combustion Institute 30 (2005) 1807–1814
Proceedings of the Combustion Institute
www.elsevier.com/locate/proci

Control of oscillating combustion and noise based on local flame structure

Gyung-Min Choi[a], Mamoru Tanahashi[b,*], Toshio Miyauchi[b]

[a] School of Mechanical Engineering, College of Engineering, Pusan National University, 30 Jangjeon-dong, Geumjeong-gu, Busan 609-735, Republic of Korea

[b] Department of Mechanical and Aerospace Engineering, Tokyo Institute of Technology, 2-12-1 Ookayama, Meguro-ku, Tokyo 152-8552, Japan

Abstract

To control combustion oscillations, the characteristics of an oscillating swirl injection premixed flame have been investigated, and control of oscillating combustion and noise based on local flame structure has been conducted. The r.m.s. value of pressure fluctuations and noise level show significantly large values between $\phi = 0.8$ and 1.1. The beating of pressure fluctuations is observed for the large oscillating flame conditions in this combustor. Relationship between beating of pressure fluctuations and local flame structure was observed by the simultaneous measurement of CH/OH planar laser induced fluorescence and pressure fluctuations. The local flame structure and beating of pressure fluctuations are related and the most complicated flame is formed in the middle pressure fluctuating region of beating. The beating of pressure fluctuations, which plays important roles in noise generation and nitric oxide emission in this combustor, could be controlled by injecting secondary fuel into the recirculating region of oscillating flames. Injecting secondary fuel prevented lean blowout, and low NO_x combustion was also achieved even for the case of pure methane injection as a secondary fuel. By injecting secondary fuel into the recirculating region near the swirl injector, the flame lifted from the swirl injector and its reaction region became uniform and widespread, hence resulting in low nitric oxide emission. Secondary mixture injection, fuel diluted with air, is not effective for control of combustion oscillations suppression and lean blowout prevention.
© 2004 Published by Elsevier Inc. on behalf of The Combustion Institute.

Keywords: Combustion control; Oscillating combustion; Local flame structure; Secondary fuel injection

1. Introduction

It has been considered that most of combustion oscillations or combustion instabilities are caused by the feedback interaction between natural acoustic modes of a combustor and oscillations of heat release rate [1]. Since such combustion oscillations or instabilities may cause harmful emissions or break down of the combustor, a number of studies have been conducted to elucidate the mechanism and the control strategies of combustion oscillation [2–11]. From the viewpoint of pollution formation, NO_x has a significant impact on the environment and studies related to NO_x reduction by passive or active control have also been conducted [12–14]. According to Rayleigh [1], we may easily control combustion oscillations by simply introducing an energy source out of phase with oscillations of heat release rate. However, it had

* Corresponding author. Fax: +81 3 5734 3982.
E-mail address: mtanahas@mech.titech.ac.jp (M. Tanahashi).

1540-7489/$ - see front matter © 2004 Published by Elsevier Inc. on behalf of The Combustion Institute.
doi:10.1016/j.proci.2004.08.249

been demonstrated that successful control strategies used variations of combusting condition or combustors [7–11]. Therefore, a detailed understanding of the combustion oscillations and/or instability mechanism is necessary for an effective and robust active combustion control.

To obtain important factors for control of combustion oscillations, several experimental studies have been conducted for flame–acoustic interactions in unstable combustors using phase-locked measurements [2,3]. Quantitative measurements of the flame's response to acoustic perturbations have also been conducted [15,16]. These measurements are heavily influenced by the combustion system. Since the heat release rate plays important roles in the sound generation mechanism in the turbulent combustion field [17], it is very informative to identify the dominant sound source through the understanding of spatial and temporal fluctuations of the heat release rate of the combustor for the development of active control schemes.

Recently, the effect of secondary fuel injection location on the effectiveness of active combustion control was studied in an atmospheric pressure using laboratory-scale dump combustor [18]. Simulation studies have been conducted to evaluate the control law under wide-range operation of a generic combustor using secondary fuel injection [19]. Combustion control using secondary fuel injection has two advantages: one is that successful combustion control is possible using considerably small amount of energy, and the second is that it reduces the actuator requirements. From these advantages, the secondary fuel injection method is desirable to suppress pressure fluctuations and to prevent lean blowout of the flame. On the other hand, because it had been considered that secondary fuel injection might produce a large amount of nitric oxide, optimum design has been required to minimize the amount of secondary fuel. Therefore, development of control scheme of secondary fuel injection with low nitric oxide is needed for effective and wide-range operation.

In this study, the relationship between pressure fluctuations and local flame structure has been investigated by simultaneous measurements of CH/OH planar laser induced fluorescence (PLIF) and pressure fluctuations to find most dominant factors that generate large pressure fluctuation and noise. CH radicals are produced at the flame front and have very narrow width enough to represent the reaction zone [20,21], where the OH prevails in the burned side [22]. Furthermore, it is well known that distribution of CH radicals agrees with that of the heat release rate [23,24]. This is why we conducted CH and OH PLIF measurement to investigate local flame structure. Control using secondary fuel injection has been conducted to suppress flame oscillating and noise and to prevent lean blowout. Variations of emission index with secondary fuel injection were also measured. The results of the present study demonstrate the relationship between local flame structure and pressure fluctuations, and successful control strategy using secondary fuel injection.

2. Swirl injection premixed combustor and experimental method

Figure 1 shows schematics of swirl injection premixed combustor and details of secondary fuel injection nozzle. This combustion rig consists of contraction section, swirl injection section, and combustion chamber. The inner diameter of 120 mm in the contraction section is reduced to 40 mm diameter in the swirl injection section. The inner cross-section of combustion chamber was 120 × 120 mm, and the outlet of combustion chamber was contracted to 60 mm × 60 mm. The total length of the combustion chamber was 590 mm. On each side of combustion chamber, a silica glass plate of 120 mm × 170 mm and 5 mm thickness was installed to allow optical access. The swirl injector has swirl vanes of 14 mm inner diameter and 40 mm outer diameter, which inclined 45° from the streamwise direction. The secondary fuel nozzle was mounted at the center of the swirl vanes, and its exit has 8 injection holes (ID = 0.6 mm) inclined 45° from streamwise direction.

The pressure transducer (TOYODA, PD104) was used for measuring pressure fluctuations in the combustion chamber. A water-cooling connection tube was used to mount the pressure transducer at the position 250 mm downstream. Sound level was measured with microphone

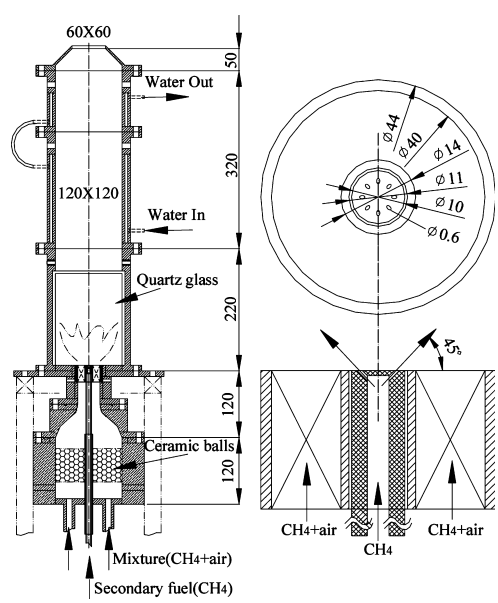

Fig. 1. Schematics of swirl injection premixed combustor and secondary fuel injection nozzle.

(ONO SOKKI, LA-1240) at the 1700 mm height from ground and 300 mm apart from the exit of combustion chamber. The concentration of nitric oxides was measured by a chemiluminescence analyzer (SHIMADZU, NOA-7000). A water-cooled sampling probe with a 1.0 mm suction hole was used to sample the burned gases at the 20 mm upstream from the exit of combustion chamber. As there are spatial fluctuations of NO_x concentration as a consequence of oscillating combustion, the averaged value of three different points was used as an emission index.

To investigate local flame structure in the oscillating flame, measurement of simultaneous CH and OH PLIF was conducted. In CH-PLIF measurement, the $Q_1(7,5)$ transition of the $B^2\Sigma^-$–$X^2\Pi(0,0)$ band at 390.30 nm was excited and fluorescence from the A–X(1,1), (0,0) and B–X(0,1) bands between 420 and 440 nm was detected [21,25]. The laser system consists of a XeCl excimer laser (Lambda Phisik, LPX 110I, 308 nm, 200 mJ/pulse) and a dye laser (Lambda Phisik, Scanmate 2) with BiBuQ dye in *p*-dioxane solvent, and generates laser pulses with about 23–25 mJ/pulse, of which pulse width is 10–20 ns. An intensified CCD camera (Andor Technology, DH734-25U-03) with 105 mm F4.5 UV lens was used for the imaging. For OH PLIF measurement, the laser system consists of an Nd-YAG laser (Spectra-Physics, Quanta Ray PRO-110, 532 nm, 350 mJ/pulse) and a dye laser (Sirah Precisionscan) with Rhodamine 590 dye in methanol solvent, and generates laser pulses with about 5 mJ/pulse at 282.93 nm, of which pulse width is 8–12 ns. The fluorescence from the A–X(1,0) and (0,0) bands ($\lambda = 306$–320 nm) was collected with a UV-Nikkor 105 mm/f4.5 lens and imaged onto a second intensified CCD camera (PI-MAX, 512RB-G1). Interference filters were used to reject incident light, KV-418 Scott for CH PLIF and WG305 and UG-11 for OH PLIF. Temporal separation of the CH and OH probe lasers was set to within 20 ns. The thickness and height of laser sheet are about 200 μm and 10 mm, respectively. The PLIF system visualized a field of 27.6 mm × 27.6 mm at a resolution of 0.027 mm/pixels for CH and 0.054 mm/pixels for OH.

The flow rates of main mixture and equivalence ratio vary from 200 to 300 L/m and from lean blowout limit to 1.3, respectively. Methane is used as fuel of main mixture and secondary fuel.

3. Combustion mode of swirl injection premixed flame

Figure 2 shows pressure fluctuations and noise as functions of flow rate of main mixture, Q_m, and equivalence ratio, ϕ. Regardless of mixture flow rates, the r.m.s. value of pressure fluctuations and noise level show significantly large values

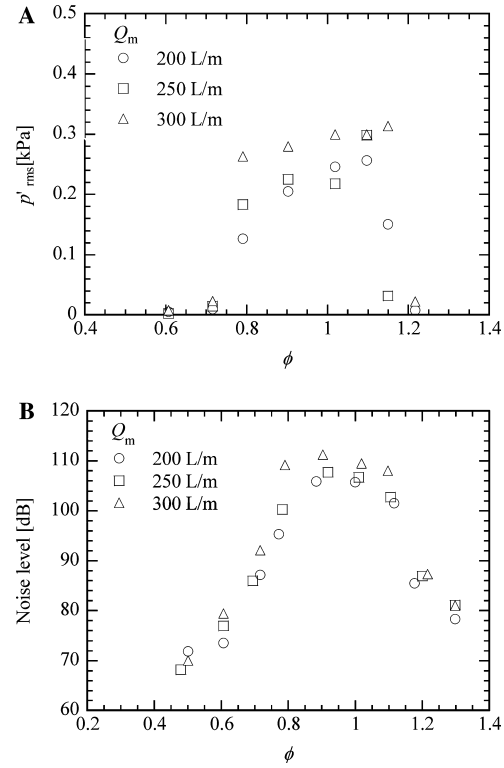

Fig. 2. R.m.s. of pressure fluctuations and noise level in the swirl injection premixed combustor.

between $\phi = 0.8$ and 1.1. The power spectrum of the pressure fluctuations at $Q_m = 300$ L/m and time-series pressure fluctuation signal at $\phi = 0.8$ are shown in Fig. 3. For the large pressure fluctuation and noisy region, most of the pressure fluctuation energy is in the 117–130 Hz range. In the present case, the dominant frequencies can be attributed to excitation of a quarter-wave mode of the combustion chamber, when we assume that the mean temperature of combustion chamber inside is 1000–1300 K. In the power spectrum of the pressure fluctuations, we can observe a second peak near the most unstable mode. Frequency of the second peak is about 30 Hz higher than that of the most unstable mode. The difference of these modes results in the beat tone as shown in Fig. 3B. We can see that the pressure signal fluctuated with frequency of the beating as well as the dominant, and the beat frequency is equal to the absolute value of the difference in frequency between the most unstable mode and neighboring one.

4. Combustion control based on local flame structure

To investigate the relationship between local flame structure and pressure fluctuations, we conducted simultaneous measurements of CH/OH

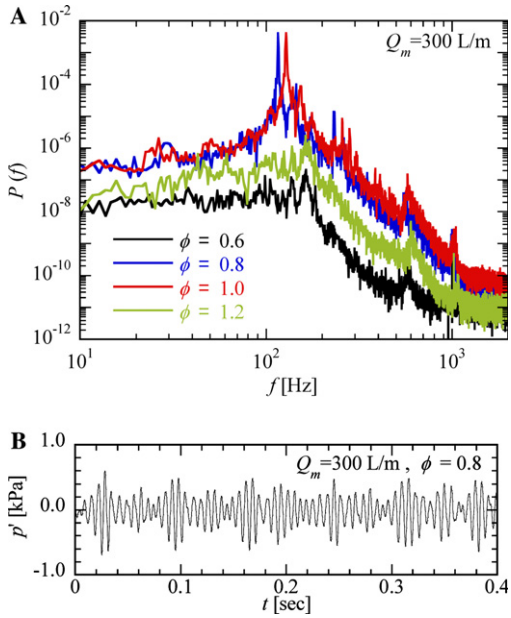

Fig. 3. Power spectrum of pressure fluctuations and time-series signal.

PLIF and pressure fluctuations, and the results are shown in Fig. 4. CH/OH PLIF measurements were conducted at maximum u'_{rms} cross-section, where u'_{rms} indicates r.m.s. value of velocity fluctuations. To investigate the turbulent characteristics of unburned mixture, two components of velocity were measured across 13 mm downstream of swirl injection exit by using a hotwire. From the simultaneous measurements of CH/OH PLIF and pressure fluctuations, we could observe a certain tendency between beating of pressure fluctuations and local flame structure. We classified the beating of pressure fluctuations into three groups, that is, small pressure fluctuating region (A), middle pressure fluctuating region (B), and large pressure fluctuating region (C) as shown schematically in Fig. 4. Representatives of CH/OH PLIF images [25 mm (W) × 9.7 mm (H)] classified into three groups are also shown in the figure. For the small pressure fluctuating region (A), the flame front is very smooth and has only slight wrinkling. The formation of flame cusps is also rare, and probability of the flame presence in the measured region seems relatively low. The scale of weak wrinkled flame front corresponds to the integral length scale (7.27 mm) of the unburned mixture. For the middle pressure fluctuating region (B), the degree of flame wrinkling and flame features were changed compared with those of the small pressure fluctuating region. The flame is much more wrinkled, and many small-scale flame cusps were observed. The scale of cusps corresponds to the order of Taylor microscale (0.949 mm) of unburned mixture, and the probability of flame presence became higher than those of the other conditions. For the large pressure fluctuating region (C), it can be seen that small-scale flame cusps remained through flame front but became rare compared to middle pressure fluctuating region. From the above results, we can see a relationship between the beating of pressure

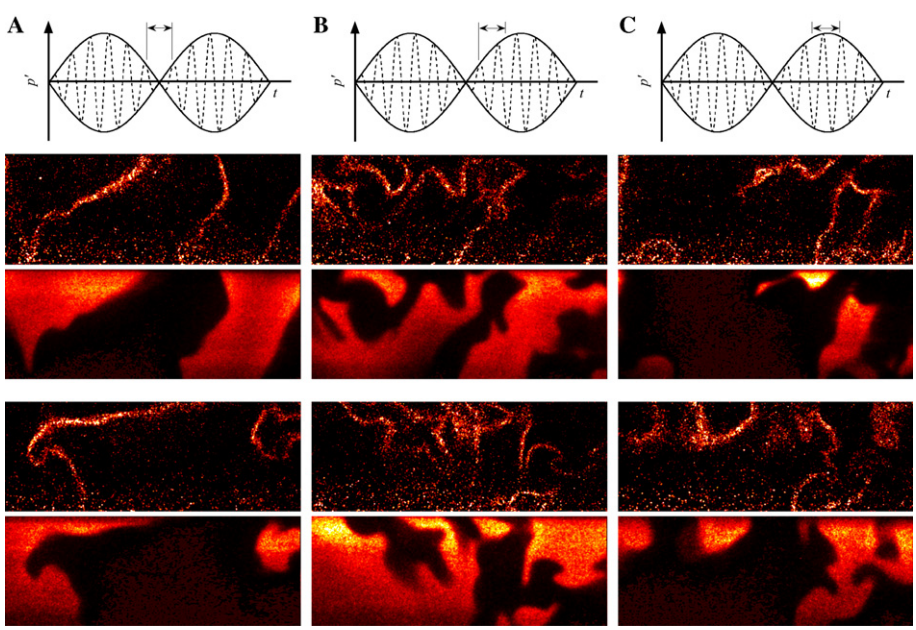

Fig. 4. CH fluorescence images (upper) and OH fluorescence images (lower) classified into three groups in the beating of pressure fluctuations ($Q_m = 300$ L/m, $\phi = 0.8$, left, A region; middle, B region; and right, C region).

fluctuations and local flame structure, and the local flame structure becomes complicated in the middle pressure fluctuating region.

Because the sound source is dominated by heat release rate in the turbulent reacting flow [17], it is thought that this complicated local flame structure plays an important role in the generation of combustion noise. To avoid the large amplitude oscillation of beating pressure fluctuations, we adopted secondary fuel injection for the control of the local flame structure. The flow rates of secondary fuel, Q_{sf}, are changed from 0.5% to 1.5% of Q_m. For cases injecting secondary fuel, the equivalence is calculated by considering both the amounts of main and secondary fuels. Figure 5 shows r.m.s. of pressure fluctuations and noise level with secondary fuel injection. From 0.5% to 1.0% of secondary fuel injection, the r.m.s. of pressure fluctuations increased with secondary fuel injection in the large oscillating regions. However, the r.m.s. of pressure fluctuations decreased drastically for the 1.5% secondary fuel injection. On the other hand, noise level decreased significantly when the secondary fuel exceeds 1.0% for the large oscillating region even though there are some increments of noise level for lean conditions. We can see that the noise level decreased over 10 dB for $\phi = 0.8$–1.0. Near $\phi = 0.8$, large noise level reduction about 20 dB is observed for 1.0% secondary fuel injection, though the r.m.s. of pressure fluctuation is larger than that without secondary fuel condition. To investigate the mechanism of noise level reduction, power spectrum of pressure fluctuations with and without secondary fuel injection was compared [in Fig. 6]. As injecting secondary fuel 1% of Q_m, the neighboring peak in the power spectrum disappeared, where the energy at the dominant frequency with secondary fuel injection is larger than that without secondary fuel [in Fig. 6A]. The case of $\phi = 0.9$ also shows this tendency. The noise level reduction in Fig. 5 is ascribed to the elimination of beating of pressure fluctuations through the control of local flame structure by injecting secondary fuel. It is thought that the loud noise is generated in the large pressure fluctuation region. This is why loud noise is generated for the case without secondary fuel injection even though the r.m.s. of pressure fluctuations without secondary fuel injection is smaller than that with secondary fuel injection. From the result of little effect in pressure fluctuations and noise level for 0.5% secondary fuel injection, it can also be seen that there is a minimum flow rate of secondary fuel to control the local flame structure in the present swirl injection premixed combustor.

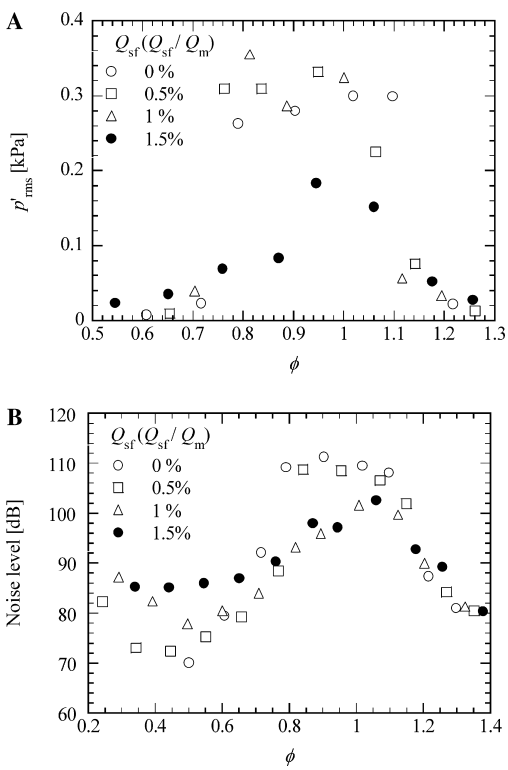

Fig. 5. Pressure fluctuations and noise level with secondary fuel injection ($Q_m = 300$ L/m).

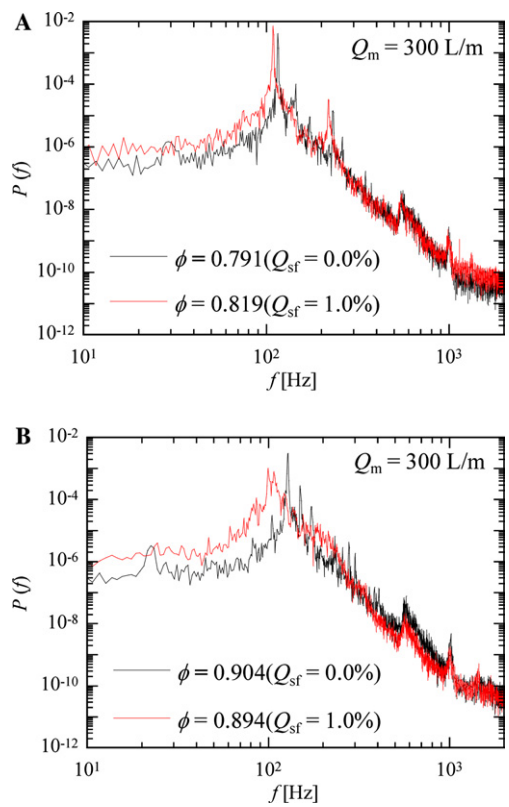

Fig. 6. Comparison of power spectrums of pressure fluctuations with and without secondary fuel injection.

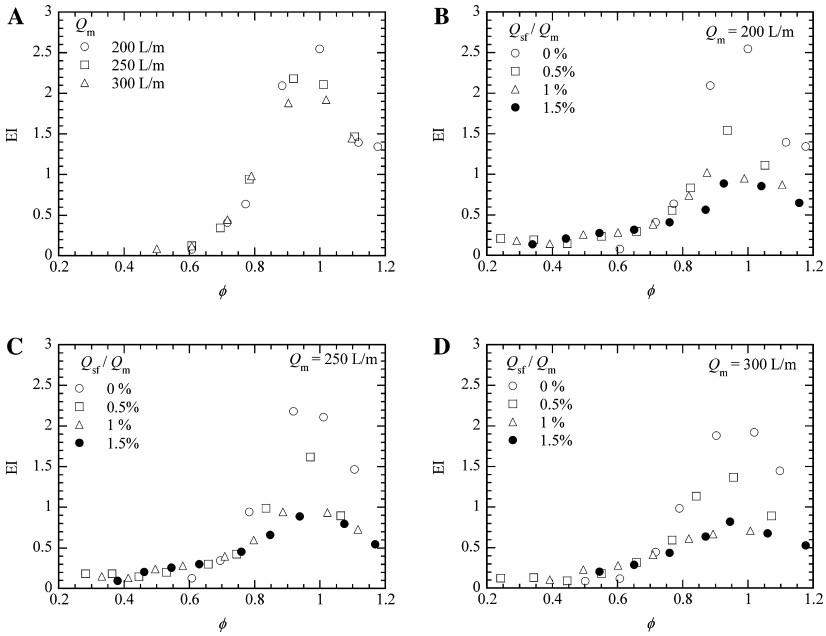

Fig. 7. Variation of emission index with secondary fuel injection.

Because pure methane is injected as a secondary fuel, if the injected pure methane reacts as a diffusion flame, a large amount of nitric oxide will be produced. Figure 7 shows the variation of emission index with secondary fuel injection. For the case without secondary fuel injection, emission index also shows a large value in the large oscillating and noisy regions for all flow rates conditions. This large amount of emission index is ascribed to the combustion oscillating with complicated flame structures. As injecting secondary fuel, we can observe that the emission index decreases in the large pressure fluctuating region. However, there is little effect of secondary fuel injection on lean and rich conditions, of which r.m.s. of pressure fluctuations are small. For $0.8 < \phi < 1.1$ and without secondary fuel injection, the flame stabilizes near the swirl injector, and locally intense reaction regions are observed by time-series CH chemiluminescence visualization, though which are not shown here. As injecting secondary fuel, the flame stabilized near the swirl injector lifts slightly, and relatively uniform and flattened flame is formed. Low emission index is ascribed to lifted, and relatively uniform and widespread flame. NO_x reduction by lifting flame from injection nozzle can also be observed for lifted flame in the highly preheated air stream [26]. But, these NO_x reduction mechanisms are different from NO destruction by fuel injection [27,28] or mild combustion [29,30]. In addition, the lean blowout limits are extended to quite lean condition by injecting secondary fuel. From the above result, it can be seen that secondary fuel injection is effective to suppress combustion oscil-

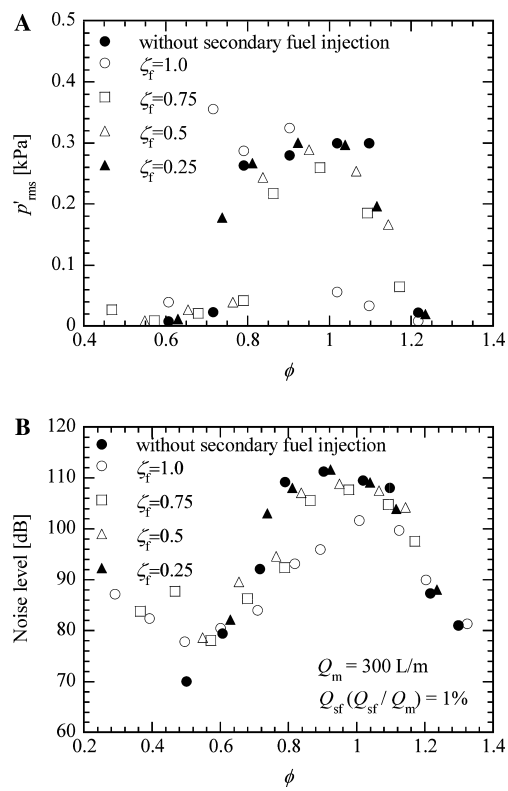

Fig. 8. Influence of secondary premixed mixture injection on the r.m.s. of pressure fluctuations and noise level.

lation, to prevent lean blowout, and to reduce emission index.

Figure 8 shows the influence of secondary premixed mixture injection on the r.m.s. of pressure fluctuations and noise with 1.0% secondary premixed mixture injection. Here, ξ_f is the ratio of fuel flow rate to mixture one. To evaluate the effect, the results without secondary fuel injection are also plotted. R.m.s. of pressure fluctuations with secondary premixed mixture injection is less than that of pure methane secondary fuel injection under $\phi = 1.0$ conditions, though opposite tendency is observed over $\phi = 1.0$ conditions. For the noise level, secondary premixed mixture injection has little effect on noise reduction. But, the noise level with secondary premixed mixture injection seldom exceeds that without secondary fuel injection. And, the lean blowout limit moved to the rich side with increasing air amounts in the secondary premixed mixture. These results demonstrate that secondary premixed mixture injection is not effective for the control of combustion oscillation suppression and lean blowout prevention.

5. Conclusions

We have carried out investigations to make clear the relationship between pressure fluctuations and local flame structure in oscillating swirl injection premixed flame, and control of local flame structure has been conducted using secondary fuel injection to suppress combustion oscillations, noise level, and nitric oxide production. Conclusions obtained by the present study include:

1. Beating of pressure fluctuations, strongly related with local flame structure, causes high level of combustion noise and a large amount of emission index in the present swirl injection premixed combustor.
2. Beating of pressure fluctuations can be controlled by secondary fuel injection, and this control is effective on suppression of combustion oscillation, prevention of lean blowout, and reduction of emission index.
3. Secondary premixed mixture injection is not effective for control of combustion oscillations suppression and lean blowout prevention.

Acknowledgments

This work was partially supported by the collaborative research project on "Smart Control of Turbulence: A Millennium Challenge for Innovative Thermal and Fluid System" of the Ministry of Education, Culture, Sports, Science and Technology.

References

[1] L. Rayleigh, *Theory of Sound*, Dover, New York, 1945.
[2] J.M. Samaniego, B. Yip, T. Poinsot, S. Candel, *Combust. Flame* 94 (1993) 363–380.
[3] J.C. Broda, S. Seo, R.J. Santoro, G. Shirhattikar, V. Yang, *Proc. Combust. Inst.* 27 (1998) 1849–1856.
[4] A. Di Benedetto, F.S. Marra, G. Russo, *Combust. Sci. Technol.* 174 (10) (2002) 1–18.
[5] C.O. Paschereit, E. Gutmark, W. Weisenstain, *Phys. Fluids* 11 (9) (1999) 2667–2678.
[6] T. Lieuwen, B.T. Zinn, AIAA paper No.2000-0707 (2000).
[7] A. Gulati, R. Mani, *J. Propul. Power* 8 (5) (1992) 1109–1115.
[8] S. Sivasegaram, R.F. Tsai, J.H. Whitelaw, *Combust. Sci. Technol.* 10 (5) (1995) 67–83.
[9] K.H. Yu, K.J. Wilson, K.C. Schadow, *Proc. Combust. Inst.* 27 (1998) 2039–2046.
[10] J.P. Hathout, M. Fleifil, A.M. Annaswamy, A.F. Ghoniem, *Proc. Combust. Inst.* 28 (2000) 721–730.
[11] R. Blonbou, A. Laverdant, S. Zaleski, P. Kuentzmann, *Proc. Combust. Inst.* 28 (2000) 747–755.
[12] C. Poppe, S. Sivaegaram, J.H. Whitelaw, *Combust. Flame* 113 (1998) 13–26.
[13] O. Delabroy, E. Haile, F. Lacas, S. Candel, A. Pollard, A. Sobiesiak, H.A. Becker, *Exp. Thermal Fluid Sci.* 16 (1998) 64–75.
[14] S. Murugappan, E.J. Gutmark, S. Acharya, M. Kristic, *Proc. Combust. Inst.* 28 (2000) 731–737.
[15] T. Poinsot, C. Le Chatelier, S.M. Candel, E.J. Esposito, *J. Sound Vibr.* 107 (2) (1986) 265–278.
[16] J. Harper, C.J. Johnson, Y. Neumeier, T. Lieuwen, B.T. Zinn, *AIAA Paper* No.2001-0486 (2001).
[17] M. Tanahashi, Y. Li, G.-M. Choi, T. Miyauchi, in: *9th Int. Conf. Num. Combust.*, 2002, pp. 201–202.
[18] J.G. Lee, K. Kim, D. Santavicca, *Proc. Combust. Inst.* 28 (2000) 739–746.
[19] B-S. Hong, A. Ray, V. Yang, *Combust. Flame* 128 (2002) 242–258.
[20] M. Allen, R.D. Howe, R.K. Hanson, *Opt. Lett.* 11 (1986) 126–128.
[21] C.D. Carter, J.M. Donbar, J.F. Driscoll, *Appl. Phys. B* 66 (1998) 129–132.
[22] M.D. Smooke, Y. Xu, R.M. Zurn, P. Lin, J.H. Frank, M.B. Long, *Proc. Combust. Inst.* 24 (1992) 813–821.
[23] M.W. Renfro, G.B. King, M. Laurendeau, *Combust. Flame* 122 (2000) 139–150.
[24] S.H. Starner, R.W. Bilger, R.W. Dibble, R.S. Barlow, D.C. Fourguette, M.B. Long, *Proc. Combust. Inst.* 24 (1992) 341–349.
[25] N.L. Garland, D.R. Crosley, *Appl. Opt.* 24 (1985) 4229–4237.
[26] T. Fujimori, D. Reichelmann, J. Sato, in: *1st Aaia-Pacific Conf. Combust.*, 1997, pp. 298–301.
[27] S.C. Lee, C.P. Koshland, D. Lucas, R.F. Sawyer, *Combust. Flame* 119 (1999) 154–160.
[28] P.G. Kristensen, P. Glarborg, K. Dam-Johansen, *Combust. Flame* 107 (1996) 211–222.
[29] G.-M. Choi, M. Katsuki, *Proc. Combust. Inst.* 29 (2002) 1165–1171.
[30] P.J. Coelho, N. Peters, *Combust. Flame* 124 (2001) 503–518.

Comments

F.C. Gouldin, Cornell University, USA. A possible mechanism for instability in your combustor is the coupling of combustor pressure oscillations with oscillations in the plenum chamber upstream of the combustor. Pressure oscillations in the plenum can cause combustor inlet velocity fluctuations influencing flame stabilization. By measuring the pressure in the plenum one can detect the presence of oscillations and their relationship to pressure fluctuations in the combustor.

Reply. In the present study, two dominant peaks can be observed in pressure fluctuation in the combustor. As you commented, these peaks may be a result of the coupling of pressure fluctuations in the combustor and in the plenum. However, we did not measure the pressure fluctuation in the plenum because our main objective was development of the control schemes based on the local flame structure, which disconnect or suppresses the coupling. We will try to measure pressure in the plenum to investigate the detailed mechanism of combustion oscillations in future work.

•

Francesco Ciucci, Caltech, USA. The evaluation of the Rayleigh Index (RI) provides good physical insight for understanding the reason why combustion instabilities appear and the regions where they appear (burner impact, flame position, etc.). Your measurement (OH, CH PLIF, and r.m.s. pressure) could have lent itself to an evaluation of the RI. You did not calculate the RI, why? Is the OH-LIF information sufficient for drawing your conclusions?

Reply. From our measurement results, we can calculate RI as you suggested. RI provides important information to specify the conditions of combustion oscillations. In the present study, however, it have been revealed that a certain relationship exists between beating of pressure fluctuations, about 30 Hz, and local flame structure for large pressure fluctuating conditions. Since the beating frequency of pressure fluctuations was relatively lower than the dominant frequency in the pressure fluctuations and the secondary fuel was injected with constant velocity, RI is not so important for controlling local flame structure. In high Reynolds number turbulence, the edge of OH does not always correspond to the flame front and CH can approximate the high heat release rate region which is the main sound source in turbulent reactive flows. Therefore, we have conducted simultaneous OH and CH PLIF.

•

Albert Ratner, University of Iowa, USA. The flame changes mode from anchored to lifted because the secondary injection occurs at the flame edge. The lifted flame can produce less NO due to several possible factors, including volumetric spread or passage of the fuel through the combustion chamber without burning. How much of the fuel is burned? Also, the injection appears to be effective, mainly, because the flame is near transition. That implies that the results are heavily geometry and flow condition dependent.

Reply. The oxygen concentration measured near the exit is almost the same value both with and without secondary fuel injection. This implies that the degrees of oxygen consumption are almost identical for both cases, although the exact unburned fuel components were not measured. The position and injection velocity of secondary fuel injection plays a decisive role to control the flame structure and to stabilize flames. Therefore, the most effective control conditions such as injection angle and injection velocity may depend on the geometry. The important point is whether the control scheme is constructed based on the local flame structure or not.

•

Robert Cheng, Lawrence Berkeley National Laboratory, USA. From your high-speed movies, it seems that the cause of your oscillations was associated with the flame lifting off and reattaching to the vanes. Can you comment if this is the main instability mechanism?

Reply. From the high-speed movies, we can observe that the frequency of flame lifting off and reattaching were almost the same with the dominant frequency in the pressure fluctuations. It is considered that lifting off and reattaching of the flame is attributed to an excitation of a quarter-wave mode of the combustor.